

EduGenie

Google Gemini Powered Learning Assistant

Project Report

Vetri Thiran Payirchi Thittam

Submitted by: _____

Register / Roll No.: _____

Institution: _____

Date: _____

1. Abstract

EduGenie is a web-based learning assistant powered by Google Gemini, a generative AI model. It helps students learn any topic through simple explanations, practice quizzes, quick summaries and personalised study plans. Learners can choose their level (Beginner, Intermediate, Advanced) and their language (English, Tamil or Tanglish), so the help feels natural and easy to follow. The application runs entirely in the browser as a single HTML file and connects directly to the Gemini API, so it needs no server and is simple to deploy and share.

2. Introduction

Many students, especially in skill-training programmes, do not get personal guidance whenever they are stuck. Books and videos are the same for everyone and cannot answer follow-up questions. Generative AI can act like a patient tutor that is available at any time. EduGenie uses this idea to give every learner a friendly, on-demand study partner.

3. Problem Statement

- Students often cannot get instant help for doubts outside class hours.
- Most study material is English-only and one-size-fits-all, which is hard for regional-language learners.
- Learners lack quick ways to self-test and to plan their preparation.
- Paid tutoring and coaching are costly for many families.

4. Objectives

- Build an AI assistant that explains topics step by step in simple words.
- Generate quizzes so learners can test themselves.
- Summarise long topics or text into key points.
- Create day-by-day study plans for a chosen goal.
- Support English, Tamil and Tanglish and three difficulty levels.
- Keep the tool free to use, lightweight and mobile-friendly.

5. Scope

EduGenie is a general-purpose learning assistant. It can be used for school subjects, technical skills (such as programming), and competitive exam preparation. It works on any device with a modern browser and an internet connection.

6. Features

Feature	Description
Explain mode	Step-by-step explanation with a real-life example and a one-line takeaway.
Quiz mode	Five multiple-choice questions with an answer key and reasons.
Summary mode	Bullet-point summary of a topic or pasted text with three key terms.
Study plan mode	Practical day-by-day plan with goals and a final self-test.
Language choice	English, Tamil (தமிழ்) or Tanglish.
Level choice	Beginner, Intermediate or Advanced.
Chat memory	Follow-up questions work because the conversation history is sent with each request.
Sample prompts	One-tap example topics for each mode.

7. Technology Stack

Component	Technology
AI model	Google Gemini (gemini-2.5-flash) via the Gemini API
Frontend	HTML5, CSS3, JavaScript (no framework)
API call	Fetch API to the generateContent endpoint
Storage	Browser localStorage (API key only)
Deployment	Single static HTML file; can be hosted on GitHub Pages or opened locally

8. System Design

8.1 Architecture

EduGenie follows a simple client-to-API architecture. The user interacts with the browser interface. The app builds a system instruction from the selected mode, level and language, adds the conversation history, and sends it to the Gemini API. The response is formatted and shown in the chat window.

8.2 Workflow

- User enters the Gemini API key once; it is saved in the browser.
- User selects a mode, language and level.
- User types a topic or question.
- The app creates a system prompt and sends the request with chat history to Gemini.

- Gemini returns the answer, which is converted to readable formatting (headings, lists, code).
- The answer is displayed and added to history for follow-up questions.

8.3 Prompt Design

Each mode has its own instruction, for example: the Quiz mode asks Gemini for five multiple-choice questions with an answer key. A common system instruction sets the tutor role, learner level, reply language and a request for short, clear answers.

9. Implementation

The whole application is one file, edugenie.html. Key parts of the code are:

- Mode prompts: a JavaScript object that stores the instruction for each learning mode.
- API request: a fetch call with the system instruction and the conversation contents.
- Markdown rendering: a small function that converts headings, bold text, lists and code blocks to HTML safely (HTML characters are escaped first).
- Error handling: API errors, an empty response or a missing key show a clear message in the chat.
- Accessibility: keyboard focus styles, labelled controls and reduced-motion support.

10. How to Run

- Download edugenie.html and open it in Chrome, Edge or any modern browser.
- Get a free API key from Google AI Studio (aistudio.google.com/apikey).
- Paste the key in the yellow box at the top and click Save key.
- Choose a mode, language and level, then type a topic and press Send.

11. Testing

Test case	Expected result	Status
Send question without API key	Message asks user to save key	Pass
Explain: "What is machine learning?"	Simple explanation with example	Pass
Quiz: "Python basics"	5 MCQs with answer key	Pass
Change language to Tamil	Reply is in Tamil	Pass
Follow-up question	Answer uses earlier context	Pass
Invalid API key	Error message shown in chat	Pass

Note: fill in the Status column after you run each test yourself.

12. Advantages

- Available 24/7 and free to run with a Gemini free-tier key.
- Supports regional language and Tanglish.
- No installation or server; works on mobile and desktop.
- Personalised to the learner's level and goal.

13. Limitations

- Needs an internet connection and a valid API key.
- AI answers can occasionally be wrong, so important facts should be verified with textbooks.
- Free-tier usage has rate limits.
- Chat history is not saved after the page is refreshed.

14. Future Enhancements

- Voice input and text-to-speech for Tamil and English.
- Upload a PDF or notes and ask questions from it.
- Save progress, quiz scores and learning history for each student.
- Interactive scored quizzes with instant feedback.
- Teacher dashboard to track class performance.

15. Conclusion

EduGenie shows how generative AI can make learning more personal, accessible and affordable. By combining Google Gemini with a simple, language-friendly interface, it gives every learner a study partner that explains, tests and plans with them. With future additions such as voice support and progress tracking, it can grow into a complete learning platform.

16. References

- Google AI Studio and Gemini API documentation – ai.google.dev
- MDN Web Docs – Fetch API and Web Storage API – developer.mozilla.org